

1 Introduction

- The **label trick** was recently proposed and has enjoyed widespread success in facilitating the parallel use of node features and labels via a stochastic label splitting technique.
- However, the label trick has been motivated primarily as a training heuristic without a strong theoretical foundation.

2 Background

- Label propagation** is a semi-supervised algorithm that predicts unlabeled nodes by propagating the observed labels across the edges of the graph, with the underlying smoothness assumption that two nodes connected by an edge are likely to share the same label.
- In contrast to the propagation of labels across the graph, **graph neural networks (GNNs)** transform and propagate node features using a series of feed-forward neural network layers.
- GNNs follow a similar propagation mechanism as the label propagation algorithm, the principal difference being whether features or labels are smoothed across the graph, it is natural to consider combinations of the two for improving performance.
- A recently emerged label trick has enjoyed widespread success in facilitating the parallel use of node features and labels via a stochastic label splitting technique, which is defined as follows.

Definition 1 (label trick) The label trick is based on creating random partitions of the training data as in $\mathcal{D}_{tr} = \mathcal{D}_{in} \cup \mathcal{D}_{out}$ and $\mathcal{D}_{in} \cap \mathcal{D}_{out} = \emptyset$, where node labels from $\mathcal{D}_{in}$ are concatenated with the original features $\mathbf{X}$ and provided as GNN inputs (for nodes not in $\mathcal{D}_{in}$ zero-padding is used), while the labels from $\mathcal{D}_{out}$ serve in the traditional role as supervision. The resulting training objective then becomes

$$\mathbb{E}_{splits} \left[\sum_{i \in \mathcal{D}_{out}} \ell(\mathbf{y}_i, f[\mathbf{X}, \mathbf{Y}_{in}; \mathbf{W}]_i) \right] \quad (2)$$

where $\mathbf{Y}_{in} \in \mathbb{R}^{n \times c}$, is defined row-wise as $\mathbf{y}_{in,i} = \begin{cases} \mathbf{y}_i & \text{if } i \in \mathcal{D}_{in} \\ \mathbf{0} & \text{otherwise} \end{cases}$ for all i , the function $f(\mathbf{X}, \mathbf{Y}_{in}; \mathbf{W})$ represents a message-passing neural network with parameters $\mathbf{W}$ and the concatenation of $\mathbf{X}$ and $\mathbf{Y}_{in}$ as inputs, and $\ell(\cdot, \cdot)$ denotes a point-wise loss function over one sample/node. At inference time, we then use the deterministic predictor $f[\mathbf{X}, \mathbf{Y}_{tr}; \mathbf{W}]_i$ for all test nodes $i \notin \mathcal{D}_{tr}$.

3 Reliable Randomness Through the Label Trick

3.1 Label Trick without Node Features

$$\mathcal{L}(\mathbf{W}) = \mathbb{E}_{splits} \left[\sum_{i \in \mathcal{D}_{out}} \ell(\mathbf{y}_i, [\mathbf{P}\mathbf{Y}_{in}\mathbf{W}]_i) \right].$$

3.2 Self-excluded Simplification of the Label Trick

The optimization objective is

$$\begin{aligned} \mathcal{L}(\mathbf{W}) &= \mathbb{E}_{splits} \left[\sum_{i \in \mathcal{D}_{out}} \ell(\mathbf{y}_i, [\mathbf{P}\mathbf{Y}_{in}\mathbf{W}]_i) \right] = \mathbb{E}_{splits} \left[\sum_{i \in \mathcal{D}_{out}} \ell(\mathbf{y}_i, [\mathbf{P}(\mathbf{Y}_{tr} - \mathbf{Y}_i)\mathbf{W}]_i) \right] \\ &= \frac{1}{|\mathcal{D}_{tr}|} \sum_{i \in \mathcal{D}_{tr}} \ell(\mathbf{y}_i, [(\mathbf{P} - \mathbf{C})\mathbf{Y}_{tr}\mathbf{W}]_i), \end{aligned} \quad (5)$$

where $\mathbf{Y}_i$ represents a matrix that shares the i -th row of $\mathbf{Y}$ and pads the rest with zeros, and $\mathbf{C} = \text{diag}(\mathbf{P})$. This then motivates the revised predictor given by

$$f(\mathbf{Y}_{tr}; \mathbf{W}) = (\mathbf{P} - \mathbf{C})\mathbf{Y}_{tr}\mathbf{W}. \quad (6)$$

3.3 Label Trick as a Regularizer

Assuming a mean square error (MSE) loss as computed for each node via $\ell(\mathbf{y}, \hat{\mathbf{y}}) = \|\mathbf{y} - \hat{\mathbf{y}}\|_2^2$ (later we consider categorical cross-entropy), our overall objective is to minimize

$$\mathcal{L}(\mathbf{W}) = \mathbb{E}_{splits} \left[\sum_{i \in \mathcal{D}_{out}} \ell(\mathbf{y}_i, [\mathbf{P}\tilde{\mathbf{Y}}_{in}\mathbf{W}]_i) \right] = \mathbb{E}_{splits} \left[\|\mathbf{Y}_{out} - \mathbf{M}_{out}\mathbf{P}\tilde{\mathbf{Y}}_{in}\mathbf{W}\|_F^2 \right], \quad (7)$$

where $\mathbf{M}_{out}$ is a diagonal mask matrix defined such that $\mathbf{Y}_{out} = \mathbf{M}_{out}\mathbf{Y} = \mathbf{Y}_{tr} - \mathbf{Y}_{in}$ and the random splits follow a node-wise Bernoulli distribution with parameter α as discussed previously. We then have the following:

Theorem 1 The label trick objective from (7) satisfies

$$\frac{1}{1 - \alpha} \mathbb{E}_{splits} \left[\|\mathbf{Y}_{out} - \mathbf{M}_{out}\mathbf{P}\tilde{\mathbf{Y}}_{in}\mathbf{W}\|_F^2 \right] = \|\mathbf{Y}_{tr} - \mathbf{M}_{tr}(\mathbf{P} - \mathbf{C})\mathbf{Y}_{tr}\mathbf{W}\|_F^2 + \frac{1 - \alpha}{\alpha} \|\mathbf{\Gamma}\mathbf{W}\|_F^2, \quad (8)$$

where $\mathbf{\Gamma} = (\text{diag}(\mathbf{P}^T\mathbf{P}) - \mathbf{C}^T\mathbf{C})^{\frac{1}{2}}\mathbf{Y}_{tr}$.

Corollary 1 Under the same conditions as Theorem 1, if we add the node feature term $\mathbf{P}\mathbf{X}\mathbf{W}_x$ to the label-based predictor from (7), we have that

$$\begin{aligned} \frac{1}{1 - \alpha} \mathbb{E}_{splits} \left[\|\mathbf{Y}_{out} - \mathbf{M}_{out}\mathbf{P}\mathbf{X}\mathbf{W}_x - \mathbf{M}_{out}\mathbf{P}\tilde{\mathbf{Y}}_{in}\mathbf{W}_y\|_F^2 \right] \\ = \|\mathbf{Y}_{tr} - \mathbf{M}_{tr}\mathbf{P}\mathbf{X}\mathbf{W}_x - \mathbf{M}_{tr}(\mathbf{P} - \mathbf{C})\mathbf{Y}_{tr}\mathbf{W}_y\|_F^2 + \frac{1 - \alpha}{\alpha} \|\mathbf{\Gamma}\mathbf{W}_y\|_F^2. \end{aligned} \quad (9)$$

Theorem 2 Under the same conditions as in Theorem 1 and Corollary 1, if we replace the MSE loss with categorical cross-entropy we obtain the bound

$$\begin{aligned} \frac{1}{1 - \alpha} \mathbb{E}_{splits} \left[\text{CrossEntropy}_{\mathcal{D}_{out}}(\mathbf{Y}_{out}, \mathbf{P}\mathbf{X}\mathbf{W}_x + \mathbf{P}\tilde{\mathbf{Y}}_{in}\mathbf{W}_y) \right] \\ \geq \text{CrossEntropy}_{\mathcal{D}_{tr}}(\mathbf{Y}_{tr}, \mathbf{P}\mathbf{X}\mathbf{W}_x + (\mathbf{P} - \mathbf{C})\mathbf{Y}_{tr}\mathbf{W}_y), \end{aligned} \quad (11)$$

where $\text{CrossEntropy}_S(\cdot, \cdot)$ denotes the sum of row-wise cross-entropy of S .

3.4 Non-linear Extensions

Theorem 3 Let $f_{GNN}(\mathbf{X}, \mathbf{Y}; \mathbf{W})$ denote an arbitrary GNN model with concatenated inputs $\mathbf{X}$ and $\mathbf{Y}$, and $\ell(\mathbf{y}, \hat{\mathbf{y}})$ a training loss such that $\sum_{i \in \mathcal{D}_{out}} \ell(\mathbf{y}_i, f_{GNN}[\mathbf{X}, \mathbf{Y}_{in}; \mathbf{W}]_i)$ is bounded for all $\mathcal{D}_{out}$. It then follows that

$$\lim_{\alpha \rightarrow 1} \left\{ \frac{1}{1 - \alpha} \mathbb{E}_{splits} \left[\sum_{i \in \mathcal{D}_{out}} \ell(\mathbf{y}_i, f_{GNN}[\mathbf{X}, \mathbf{Y}_{in}; \mathbf{W}]_i) \right] \right\} = \sum_{i=1}^m \ell(\mathbf{y}_i, f_{GNN}[\mathbf{X}, \mathbf{Y}_{tr} - \mathbf{Y}_i; \mathbf{W}]_i). \quad (12)$$

4 Broader Use Cases of the Label Trick

4.1 Trainable Label Propagation

The label trick can be adopted to learn a label propagation weight matrix. Other entry points are also feasible such as node-dependent

weights, nonlinear weighted mappings, step-wise weights for heterophily graphs, or weights for different node types, etc.

4.2 Deterministic Label Trick

The label trick can be deterministically applied to GNNs with linear propagation layers, with the predictor

$$h_1 \left(\sum_i [\mathcal{P}h_0([\mathbf{X}, \mathbf{Y} - \mathbf{Y}_i])]_i \right) = h_1(\mathcal{P}h_0([\mathbf{X}, \mathbf{Y}]) - \mathcal{C}h_0([\mathbf{X}, \mathbf{Y}]) + \mathcal{C}h_0([\mathbf{X}, \mathbf{0}]]), \quad (13)$$

where $[\cdot, \cdot]$ denotes the concatenation operation, $\mathcal{P} = [\mathcal{P}_0, \mathcal{P}_1, \dots, \mathcal{P}_{k-1}]^T$ is the integrated propagation matrix, $\mathcal{C} = [\text{diag}(\mathcal{P}_0), \text{diag}(\mathcal{P}_1), \dots, \text{diag}(\mathcal{P}_{k-1})]^T$, h_0 and h_1 can be arbitrary node-independent functions, typically multi-layer perceptrons (MLPs).

4.3 Trainable Correct and Smooth

With the label trick, we can equip C&S with trainable weights to further boost performance. For each split the predictor is

$$\begin{aligned} f_{TC\&S}(\mathbf{Y}_{in}, \tilde{\mathbf{Y}}; \mathbf{W}) &= \mathbf{P}_s(\mathbf{Y}_{in} + (\mathbf{M}_{te} + \mathbf{M}_{out})(\tilde{\mathbf{Y}} + \tilde{\mathbf{E}}_{in}\mathbf{W}_c))\mathbf{W}_s \\ &= \mathbf{P}_s((\mathbf{Y}_{in} + (\mathbf{M}_{te} + \mathbf{M}_{out})\tilde{\mathbf{Y}})\mathbf{W}_s + \mathbf{P}_s(\mathbf{M}_{te} + \mathbf{M}_{out})\tilde{\mathbf{E}}_{in}\mathbf{W}_c\mathbf{W}_s) \\ &= \mathbf{P}_s((\mathbf{Y}_{in} + (\mathbf{M}_{te} + \mathbf{M}_{out})\tilde{\mathbf{Y}})\mathbf{W}_s + \mathbf{P}_s(\mathbf{M}_{te} + \mathbf{M}_{out})\tilde{\mathbf{E}}_{in}\tilde{\mathbf{W}}_c), \end{aligned} \quad (14)$$

where $\mathbf{W} = \{\tilde{\mathbf{W}}_c, \mathbf{W}_s\}$ and $\tilde{\mathbf{W}}_c$ is the reparameterization of $\mathbf{W}_c\mathbf{W}_s$. The objective function for optimizing $\mathbf{W}$ is

$$\mathcal{L}(\mathbf{W}) = \mathbb{E}_{splits} \left[\sum_{i \in \mathcal{D}_{out}} \ell(\mathbf{y}_i, f_{TC\&S}[\mathbf{Y}_{in}, \tilde{\mathbf{Y}}; \mathbf{W}]_i) \right]. \quad (15)$$

5 Experiments

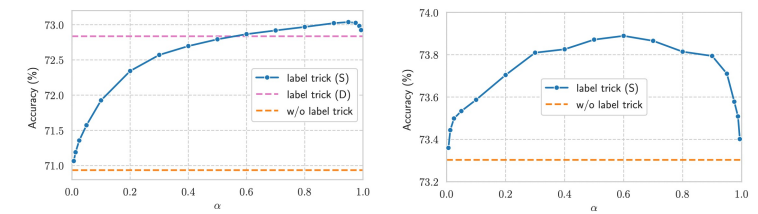


Figure 1. Accuracy results on validation dataset of linear propagation of features and labels (left) and GNNs (right) varying α .

Table 1. Accuracy results (%) of label propagation and trainable label propagation.

Method	Label Propagation	Trainable Label Propagation
Cora-full	66.44 ± 0.93	67.23 ± 0.66
Pubmed	83.45 ± 0.63	83.52 ± 0.59
Arxiv	67.11 ± 0.00	68.42 ± 0.01
Products	74.24 ± 0.00	75.61 ± 0.21

Table 2. MSE on node regression tasks with GBDT base model and C&S.

Method	C&S	Trainable C&S
House	0.51 ± 0.01	0.45 ± 0.01
County	1.42 ± 0.14	1.13 ± 0.09
VK	7.02 ± 0.20	6.95 ± 0.22
Avazu	0.106 ± 0.014	0.106 ± 0.014

Table 3. Accuracy results (%) with/without label trick (D).

Method	label trick (D)	SGC		SIGN		TWIRLS	
		×	✓	×	✓	×	✓
Cora-full		65.87 ± 0.61	65.81 ± 0.69	68.54 ± 0.76	68.44 ± 0.88	70.36 ± 0.51	70.40 ± 0.71
Pubmed		85.02 ± 0.43	85.23 ± 0.57	87.94 ± 0.52	88.09 ± 0.59	89.81 ± 0.56	90.08 ± 0.52
Arxiv		69.07 ± 0.01	70.22 ± 0.03	69.97 ± 0.16	70.98 ± 0.21	72.93 ± 0.19	73.22 ± 0.10